Notes: Faulkender and Petersen (RFS, 2006)
Does the Source of Capital Affect Capital Structure?

Contents

1 Big picture

- **Question.** Does a firm's source of debt capital affect its capital structure, above and beyond standard firm-level determinants of leverage?
- **Core claim.** Capital structure is not only a demand-side choice based on tax shields, bankruptcy costs, investment opportunities, and asset characteristics. It also reflects the supply of debt capital available to the firm.
- **Main proxy.** The paper uses whether a firm has a public debt rating—bond rating or commercial paper rating—as a proxy for access to public debt markets.
- **Main empirical fact.** Firms with a debt rating have much higher leverage than firms without a rating. In raw data, market leverage is 28.4% for firms with access versus 17.9% for firms without access.
- **Main regression result.** After controlling for firm characteristics that standard capital structure theory says affect demand for debt, rated firms remain about 7–8 percentage points more levered in market-value debt ratios.
- **Endogeneity concern.** Debt ratings are not randomly assigned. Firms may obtain ratings because they want more debt or because they have unobservable debt demand.
- **IV strategy.** The paper instruments for access using variables related to the likelihood of being rated but plausibly not directly related to leverage demand, including S&P 500 membership, NYSE listing, industry rating prevalence, firm age, and a bond-issue-size proxy.
- **IV result.** Instrumenting still leaves a large access effect: having a rating raises leverage by about 5.7–6.3 percentage points in second-stage estimates.
- **Interpretation.** Firms without public debt market access appear debt constrained or underlevered relative to otherwise similar firms with access.
- **Takeaway.** Empirical capital structure models should include supply-side access to capital markets, not only firm characteristics that determine desired leverage.

2 Incremental contribution of the paper

- **Supply-side capital structure contribution.** Earlier empirical capital structure work treated leverage as a function of firms' demand for debt. This paper explicitly adds the supply of debt capital to the empirical model.
- **Public bond market access as an observable margin.** The paper operationalizes capital-market access using public debt ratings, making a broad debt-supply channel measurable in Compustat-scale data.
- **Separates lender choice from leverage choice.** The paper connects the banking/intermediation literature on relationship versus arm's-length lending to the corporate leverage literature.
- **Shows robustness beyond firm characteristics.** The access effect remains after controlling for size, age, profitability, tangibility, market-to-book, R&D, advertising, taxes, stock returns, asset volatility, debt maturity, industry effects, and firm fixed effects.
- **Introduces an IV approach to bond-market access.** The paper uses first-stage predictors of rating access to address the concern that rating status is endogenous to debt demand.
- **Clarifies credit constraints versus capital constraints.** The evidence is strongest that non-rated firms are constrained in debt markets, while evidence that they are fully capital constrained is weaker.
- **Provides a practical empirical control.** The paper establishes debt-rating status as a useful proxy for access to public debt and for debt-market constraints in later corporate finance work.

3 Data and measurement

3.1 Sample construction

- The sample comes from Compustat for 1986–2000, using the industrial/full coverage and research files.
- The authors exclude financial firms, public-sector firms, and observations with sales or assets below $1 million.
- The main panel contains public firms, which makes the result more striking because public firms should be less financially opaque than private firms.
- In Table 1 Panel A, the sample has 77,659 firm-year observations, of which 19.0% have a debt rating.
- In the positive-debt sample, there are 69,589 firm-years, and 21.1% have a debt rating.

Interpretation: The sample is not focused on small private firms. The paper shows that even among publicly traded firms, access to public debt markets is limited and associated with leverage.

3.2 Leverage measures

- Debt includes both short-term debt and long-term debt, including the current portion of long-term debt.
- The main dependent variable is total debt divided by the market value of assets.
- Market value of assets is book assets minus book equity plus market equity.
- Robustness tests use book-value leverage, net debt, debt plus accounts payable, debt plus operating leases, and interest coverage.

Interpretation: Using both market and book leverage helps separate valuation effects from accounting capital structure. Interest coverage provides a non-ratio alternative that captures the burden of debt service.

3.3 Bond market access

- A firm has bond market access if it reports either a bond rating or a commercial paper rating.
- This is a proxy: the authors do not directly observe every debt issue's public/private source.
- The paper argues that the proxy is informative because firms with public debt almost always have a rating in other hand-collected samples.
- Misclassification likely biases the estimated access effect toward zero because some firms classified as no-access may in fact have access.

Interpretation: The rating variable is both a strength and a limitation. It is available for a large panel, but the paper must show that it captures supply-side access rather than unobserved demand for debt.

3.4 Control variables

The main regressions control for standard debt-demand determinants:

- firm size: log market value of assets;
- firm age: log of one plus age;
- profitability: profits to sales;

- tangibility: tangible assets to assets;
- growth opportunities: market-to-book assets;
- R&D to sales;
- advertising to sales;
- marginal tax rate;
- prior stock return;
- asset volatility;
- debt maturity composition.

Interpretation: These controls are chosen to absorb demand-side motives for debt. If rating status still matters after these controls, it is more plausible that ratings capture supply-side access.

4 Empirical specifications

4.1 Supply and demand framework

The paper begins from debt demand and debt supply:

$$Q^{demand} = \alpha_0 Price + \alpha_1 X_{demand} + \varepsilon_{demand},$$
$$Q^{supply} = \beta_0 Price + \beta_1 X_{supply} + \varepsilon_{supply}.$$

Solving for observed debt gives the reduced form:

$$Q^{observed} = \gamma_D X_{demand} + \gamma_S X_{supply} + \eta = \gamma_D X_{demand} + \gamma_S BondMarketAccess + \eta.$$

Interpretation: This specification is the core conceptual move. The paper argues that capital structure regressions omit an important supply shifter if they include only demand-side firm characteristics.

4.2 Baseline leverage regression

$$DebtRatio_{it} = \alpha + \beta Rating_{it} + X_{it}'\Gamma + \lambda_t + \varepsilon_{it}.$$

The coefficient β measures how much more levered firms with public debt market access are relative to firms without access, conditional on observed debt-demand determinants.

Interpretation: A positive β can be interpreted as a supply effect only if the controls absorb demand-side differences or if rating status is instrumented.

4.3 Panel-data specifications

The paper estimates several panel variants:

- industry fixed effects / within-industry regressions;
- between-industry regressions;
- firm fixed effects / within-firm regressions;
- between-firm regressions;
- first-difference regressions.

$$DebtRatio_{it} = \alpha_i + \lambda_t + \beta Rating_{it} + X_{it}'\Gamma + \varepsilon_{it}.$$

Interpretation: Firm fixed effects compare the same firm when rated and unrated, absorbing constant unobserved demand for debt. The coefficient falls but remains positive.

4.4 Interest-coverage robustness

$$\log(1 + Coverage_{it}) = \alpha + \beta Rating_{it} + X'_{it}\Gamma + \lambda_t + \varepsilon_{it}.$$

Since higher debt burdens lower coverage, a negative coefficient on *Rating* supports the same conclusion as a positive coefficient in leverage regressions.

Interpretation: The interest-coverage test addresses the concern that debt-to-assets ratios miss the burden of debt when expected cash flows differ.

4.5 First-stage probit for bond market access

$$Pr(Rating_{it} = 1) = \Phi(Z'_{it}\Pi + X'_{it}\Gamma + \lambda_t),$$

where instruments include S&P 500 membership, NYSE listing, industry rating prevalence, industry rating prevalence weighted by market value, young-firm status, and a minimum-bond-issue-size proxy.

Interpretation: The first stage attempts to distinguish firms that cannot get a rating from firms that could get a rating but choose not to issue rated debt.

4.6 Second-stage IV leverage regression

$$DebtRatio_{it} = \alpha + \beta\widehat{Rating}_{it} + X'_{it}\Gamma + \lambda_t + \varepsilon_{it}.$$

Interpretation: The second stage asks whether predicted access to the public debt market, not merely observed rating choice, raises leverage.

5 Identification strategy

5.1 Core identification problem

Having a rating is endogenous because access and leverage are jointly chosen. A firm might have high leverage because it has access to public debt markets, or it might obtain a rating because it wants to issue more debt. The authors address this problem in three layers.

5.2 Layer 1: demand-side controls

The first layer controls for firm characteristics associated with desired leverage. If rated firms remain more levered after these controls, then rating status is unlikely to be only a proxy for tax benefits, tangibility, profitability, size, or investment opportunities.

5.3 Layer 2: fixed effects and time variation

Industry and firm fixed effects absorb persistent unobservable demand for debt. The firm-fixed-effect estimate uses only firms that change rating status. A positive coefficient in this design suggests that access matters when the same firm moves into or out of the rated-debt market.

5.4 Layer 3: instrumental variables

The IV strategy predicts bond-market access with variables that affect access but plausibly do not directly shift desired leverage after controls:

- S&P 500 membership raises visibility and demand for rated securities;
- NYSE listing raises visibility and information production;
- industry rating prevalence measures rating-market infrastructure and comparability;
- young-firm status captures limited credit history;
- minimum-issue-size proxy captures fixed costs of public bond issuance.

Interpretation: The key exclusion restriction is that these variables affect leverage through public debt access, not directly through unobserved desired leverage. This assumption is more plausible for visibility and issue-size variables than for variables that also proxy for firm quality.

5.5 Remaining limitations

- Debt rating is an imperfect proxy for actual public debt use.
- IV validity depends on exclusion restrictions that cannot be proven directly.
- The paper cannot fully decompose quantity-channel and price-channel effects.
- The analysis is limited to public firms with Compustat data.

Interpretation: Despite these limitations, the consistency of raw differences, controls, fixed effects, interest coverage, and IV estimates makes the supply-side interpretation compelling.

6 Tables and figures

6.1 Figure 1 and Figure 2: Conceptual path to rating and rarity of rated debt

Read: Figure 1 distinguishes whether a firm can get a rating, chooses to get rated, chooses leverage, and receives an actual rating. Figure 2 shows that only about 19–22% of firms have ratings each year, yet rated firms account for roughly 70–86% of outstanding debt.

Interpretation: Figure 1 frames rating status as an imperfect proxy for access: the ideal test would observe whether a firm can get a rating, but the actual data observe whether it has a rating. Figure 2 shows why this matters: public debt is uncommon by firm count but dominant by debt volume.

Economic message: Public bond markets are large in dollars but accessible mainly to a minority of firms. Capital structure regressions that ignore access confound desired debt with available debt.

not want a rating. Such an approach ensures that we are capturing a

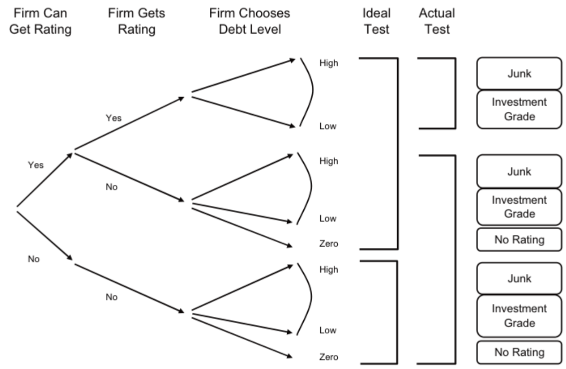

Figure 1
Bond market access, bond rating, and leverage
The figure describes the path of decisions available to the firm. First, the firm either has access to the bond market or it does not. We cannot directly observe this classification. Firms that have access to the bond market then choose whether to get a bond rating and issue public bonds. Then, conditional on their bond market access and whether they have chosen to issue public bonds, they choose their leverage. Finally, based on the firm's leverage and characteristics, the firm receives a debt rating, but only if it has issued public bonds [see Molina (2004), for an empirical test of this relationship]. We have diagramed the rating a firm could expect if it does not issue public bonds, but this hypothetical rating is

Figure 1: Bond market access, bond rating, and leverage.

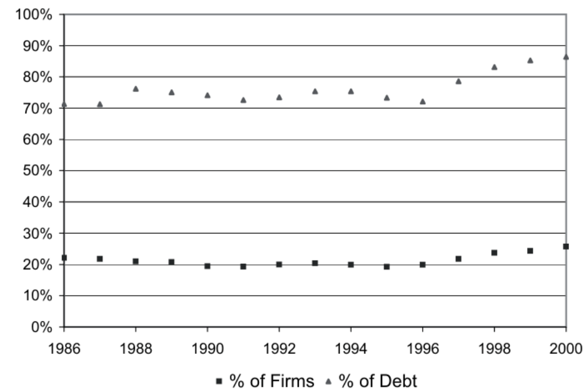

Figure 2
Percent of firms or debt with a debt rating
The figure shows the percent of firms with a debt rating (squares) or the percent of outstanding debt (in dollars) issued by firms with a debt rating (triangles). A firm has a debt rating if it reports either a bond rating or a commercial paper rating. The sample is based on firms from Compustat that report sales and assets above $1 million between 1986 and 2000 and only includes firm-years with debt.

Figure 2: Percent of firms or debt with a debt rating.

6.2 Table 1 and Table 2: Leverage and firm characteristics by access

Read: Table 1 shows rated firms have market leverage of 28.4% versus 17.9% for unrated firms, a 10.5 percentage-point difference. Median market leverage differs by 13.7 percentage points. Table 2 shows rated firms are larger, older, more profitable, more tangible, have lower market-to-book, lower R&D/sales, lower volatility, and higher tax rates.

Interpretation: Table 1 establishes the raw leverage gap. Table 2 shows why raw differences are not enough: rated firms differ systematically in ways that also affect leverage demand.

Economic message: The paper must show that access matters conditional on firm characteristics, because rated firms look like firms that standard tradeoff theory would predict to have more debt.

Table 1
Leverage by bond market access

	Mean	25%	Median	75%
Panel A: All firm-years				
Debt/asset (MV)				
Total Sample	19.9	3.0	15.3	31.7
Bond market access	28.4	14.8	25.7	38.4
No access	17.9	1.6	12.0	29.1
Difference	10.5*		13.7*	
Debt/asset (BV)				
Total sample	26.1	6.2	23.1	39.4
Bond market access	37.2	23.9	34.5	46.8
No access	23.5	3.4	18.8	36.8
Difference	13.8*		15.7*	
Panel B: Firm-years with positive debt				
Debt/asset (MV)				
Total sample	22.2	6.3	18.3	33.7
Bond market access	28.5	14.9	25.8	38.5
No access	20.5	4.6	15.7	31.9
Difference	8.0*		10.1*	
Debt/asset (BV)				
Total sample	29.1	11.5	26.5	41.4
Bond market access	37.4	24.0	34.6	46.8
No access	26.9	8.7	23.2	39.5
Difference	10.5*		11.4*	

The table reports summary statistics on firms' total debt ratios by whether they have access to the public debt markets. We use whether the firm has a debt rating as a measure of whether it has access to the public debt markets. The market value (MV) ratio is total (short- and long-term) debt divided by the book value (BV) of assets minus the BV of equity plus the MV of equity. The BV debt ratio is debt divided by the BV of assets. The BV ratio is not always between 0 and 1; it is above 1 for 1.3% of the sample. We re-coded the BV ratio to 1 for these observations. The table reports the mean, the 25th, 50th (median), and 75th percentile in each cell, except for the difference row. This row contains the difference in the means (or medians) and the associated significance levels (* indicates statistical significance at 1% level). In panel A there are 77,659 firm-year observations of which 19.0% have a debt rating. Panel B contains only firm-years in which the firm had a positive amount of debt. In panel B, there are 69,589 firm-year

Table 1: Leverage by bond market access.

Table 2
Summary statistics of firm characteristics

	Access	No access	Difference
Log(MV of assets)	7.74	4.56	3.18*
	(7.69)	(4.47)	(3.22*)
Log(BV of assets)	7.41	4.11	3.30*
	(7.34)	(4.06)	(3.29*)
Log of sales	7.21	4.11	3.10*
	(7.22)	(4.10)	(3.12*)
Log(1 + firm age)	2.61	1.83	0.78*
	(2.89)	(1.95)	(0.94*)
Profit margin (%)	16.23	2.4	13.83*
	(14.51)	(8.08)	(6.43*)
Plant, property, and equipment/assets (BV) (%)	42.39	30.84	11.55*
	(38.63)	(24.35)	(14.28*)
MV of assets/BV of assets (%)	1.59	1.88	–0.30*
	(1.30)	(1.36)	(–0.06*)
R&D/sales (%)	1.77	6.11	–4.34*
	(0.00)	(0.00)	(0.00)
Advertising/sales (%)	1.11	1.31	–0.20*
	(0.00)	(0.00)	(0.00)
Marginal tax rate (%) (before interest expense)	32.61	26.46	6.15*
	(34.99)	(34.00)	(1.00*)
Equity return previous year (%)	13.35	10.97	2.38*
	(9.02)	(–1.33)	(10.35*)
Implied asset volatility (%)	18.89	40.73	–21.84*
	(16.13)	(34.19)	(–18.06*)

The table contains summary statistics for the sample of firms with and without access to the public debt

Table 2: Summary statistics of firm characteristics.

6.3 Table 3 and Table 4: Debt maturity and determinants of market leverage

Read: Table 3 shows rated firms use much longer-maturity debt: 58.5% of rated-firm debt matures after five years versus 28.3% for unrated firms. Table 4 shows that, after controls, rating status adds about 7.1–8.3 percentage points to market leverage.

Interpretation: Table 3 shows that rated firms differ not only in debt level but also in debt structure. Table 4 shows the main result: after controlling for standard leverage determinants, rated firms remain substantially more levered.

Economic message: Public debt access appears to relax debt constraints or lower the effective cost of debt, not merely select firms that want debt.

percentiles. The sample is based on firms from Compustat that report sales and assets above $1 million between 1986 and 2000 and only includes firm-years with debt. R&D, research and development.

Table 3
Maturity of debt by bond market access

	1	2	3	4	5	>5
Total sample	32.7 (20.5)	11.6 (4.6)	8.8 (3.2)	6.6 (1.6)	5.6 (0.4)	34.8 (24.6)
Bond market access	16.4 (8.8)	5.7 (2.4)	6.1 (2.5)	6.4 (2.4)	6.9 (2.2)	58.5 (61.6)
No access	37.1 (26.2)	13.2 (5.8)	9.5 (3.6)	6.6 (1.3)	5.3 (0.1)	28.3 (11.4)
Difference	−20.7* (−17.4*)	−7.5* (−3.3*)	−3.4* (−1.2*)	−0.2** (−1.0*)	1.7* (2.1*)	30.1* (50.3*)

The table reports the mean (median) fraction of outstanding debt by maturity. Firms that have a debt rating are classified as having access. The first five columns contain the fraction of debt due in years 1 through five. The final column contains the fraction of debt with a maturity of greater than five years. The debt due in one year includes both debt with an initial maturity of less than one year and the current portion of long-term debt. The last row contains the difference in the means (or medians) between firms with and without bond market access (a debt rating). The associated significance levels also are reported (* indicates statistical significance at 1% level; ** indicates statistical significance at 5% level). The sample is based on firms from Compustat that report sales and assets above $1 million between 1986 and 2000 and only includes firm-years with debt.

Table 3: Maturity of debt by bond market access.

Table 4
Determinants of market leverage: firm characteristics

	I	II	III	IV	V
Firm has a debt rating (1 = yes)	0.083* (0.005)	0.080* (0.006)	0.078* (0.005)	0.078* (0.004)	0.071* (0.004)
Ln(market assets)	−0.010* (0.001)	−0.010* (0.001)	−0.007* (0.001)	−0.025* (0.001)	−0.026* (0.001)
Ln(1 + firm age)	−0.007* (0.001)	−0.007* (0.001)	−0.014* (0.002)	−0.016* (0.001)	−0.016* (0.001)
Profits/sales	−0.067* (0.007)	−0.075* (0.009)	−0.056* (0.009)	−0.073* (0.006)	−0.074* (0.006)
Tangible assets	0.151* (0.008)	0.151* (0.007)	0.131* (0.009)	0.129* (0.007)	0.116* (0.007)
Market-to-book (assets)	−0.040* (0.001)	−0.044* (0.002)	−0.047* (0.001)	−0.020* (0.001)	−0.019* (0.001)
R&D/sales	−0.180* (0.009)	−0.195* (0.014)	−0.198* (0.011)	−0.079* (0.007)	−0.083* (0.007)
Advertising/sales	−0.116* (0.024)	−0.070** (0.024)	−0.071 (0.045)	−0.036*** (0.022)	−0.036*** (0.021)
Marginal tax rate			−0.139* (0.014)		
Stock return previous year				−0.006* (0.001)	−0.008* (0.001)
σ (asset return)				−0.334* (0.009)	−0.326* (0.009)
% of debt due in ≤ 1 year					−0.027* (0.004)
% of debt due in >5 years					0.026* (0.004)
Number of observations	63,272	63,272	48,021	59,562	59,562
R^2	0.242	0.230	0.242	0.368	0.373

The dependent variable is the ratio of total debt to the market value (MV) of the firm's assets. White heteroscedastic consistent errors, corrected for correlation across observations of a given firm, are reported in parentheses [Rogers (1993) and White (1980)] except in column II. In column II, the coefficients and standard errors are estimated using the Fama–MacBeth method (1973). The MV of assets is the book value (BV) of assets minus the BV of equity plus the MV of debt. All models also include year dummy variables and a dummy variable for the regulated utility industry (4900–4939). The sample is based on firms from Compustat that report sales and assets above $1 million between 1986 and 2000 and only includes firm-years with debt. * indicates statistical significance at 1% level; ** indicates statistical significance at 5% level; ***

Table 4: Determinants of market leverage: firm characteristics.

6.4 Figure 3 and Figure 4: Functional form and time variation

Read: Figure 3 shows that the relation between size and predicted leverage is approximately linear, validating the log-size specification. Figure 4 plots the yearly coefficient on rating from Table 4-style regressions; it is always positive and ranges from about 5.3% to 8.6%.

Interpretation: These figures address two robustness concerns. Figure 3 shows the access effect is not a disguised nonlinear size effect. Figure 4 shows the access effect is not driven by one year or one episode.

Economic message: The supply-access premium in leverage is stable across the sample period and is not a mechanical artifact of size functional form.

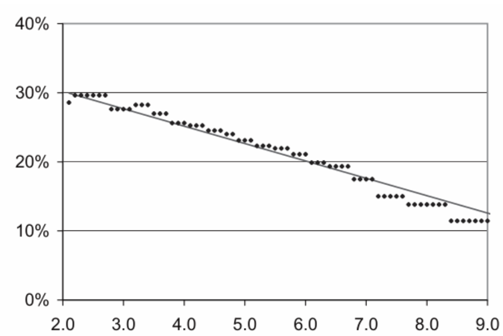

Figure 3
Effect of firm size on leverage: semi-parametric approach
The figure graphs predicted leverage as a function of firm size [log of the market value (MV) of a
The straight line is the predicted leverage based on the coefficient estimates in Table 4, column I
then estimate a second model in which the log of the MV of assets is replaced by 20 dummy variabl
for each of 20 vigintiles based on the firm's MV of assets. The diamonds graph the predicted levera
function of firm size based on the second model. Allowing for a non-linear relationship between le
and size raises the explanatory power of the model trivially (the R^2 rises from 0.368 to 0.371).

Figure 3: Effect of firm size on leverage: semiparametric approach.

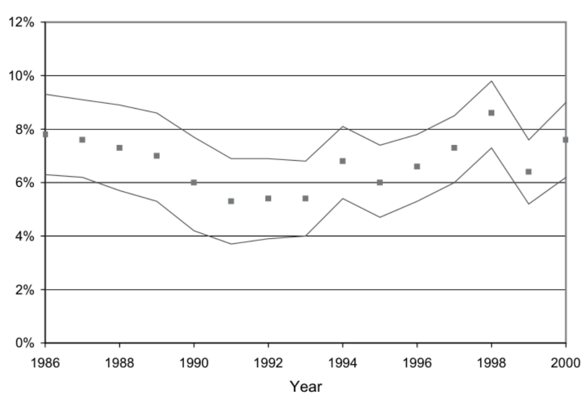

Figure 4
Effect of rating on leverage: time variation
The figure contains the estimated coefficients from a regression of leverage on the firm having a rating,
where a separate coefficient is estimated for each year. The regression includes the same controls as are
reported in Table 4, column IV. The lines denote the 99% confidence interval around the estimates.

Figure 4: Effect of rating on leverage: time variation.

6.5 Table 5 and Figure 5: Panel estimates and interpretation of within/first-difference results

Read: Table 5 reports industry and firm panel specifications. The within-firm coefficient on rating remains about 5.1%; first-difference coefficients are smaller. Figure 5 explains mechanically why first-difference estimates can be small if firms adjust slowly after obtaining a rating.

Interpretation: Table 5 is the most important control for fixed unobservable demand factors. Figure 5 clarifies that a small first-difference coefficient does not necessarily mean the access effect is small; it can reflect gradual adjustment toward a desired debt ratio.

Economic message: The same firm becomes more levered when it gains rated-debt access, supporting a supply-side interpretation.

Table 5
Determinants of market leverage: panel data estimation

	I	II	III	IV	V
Firm has a debt rating (1 = yes)	0.068* (0.002)	0.139* (0.030)	0.051* (0.002)	0.090* (0.005)	0.041* (0.003)
Ln(market assets)	−0.025* (0.000)	−0.041* (0.006)	−0.006* (0.001)	−0.022* (0.001)	0.011* (0.002)
Ln(1 + firm age)	−0.009* (0.001)	−0.077* (0.008)	.037* (0.001)	−0.023* (0.002)	0.041* (0.002)
Profits/sales	−0.072* (0.003)	−0.002 (0.046)	−0.055* (0.004)	−0.073* (0.006)	−0.044* (0.004)
Tangible assets	0.122* (0.004)	0.097* (0.022)	0.160* (0.006)	0.130* (0.006)	0.119* (0.009)
Market-to-book (assets)	−0.020* (0.001)	−0.004 (0.009)	−0.018* (0.001)	−0.016* (0.001)	−0.015* (0.001)
R&D/sales	−0.053* (0.006)	.178** (0.076)	−0.047* (0.009)	−0.060* (0.009)	−0.028* (0.008)
Advertising/sales	−0.034** (0.014)	−0.145 (0.186)	−0.033 (0.020)	−0.026 (0.020)	−0.018 (0.021)
Stock return previous year	−0.008* (0.001)	−0.072*** (0.039)	−0.017* (0.001)	−0.005 (0.004)	−0.020* (0.001)
(asset return)	−0.311* (0.003)	−0.682* (0.054)	−0.232* (0.003)	−0.359* (0.007)	−0.185* (0.004)
Number of observations	59,562	59,562	59,562	59,562	49,742
R^2	0.442	0.612	0.763	0.465	0.272
Controls	Industry	Industry	Firm	Firm	Firm
Estimation method	Within	Between	Within	Between	Changes

The dependent variable is the ratio of total debt to the market value (MV) of the firm's assets. The MV of assets is the book value (BV) of assets minus the BV of equity plus the MV of debt. All models also include year dummy variables and a dummy variable for the regulated utility industry (4900–4939). The sample is based on firms from Compustat that report sales and assets above $1 million between 1986 and 2000 and only includes firm-years with debt. Column I—within industry estimates. The coefficients are estimated based on variation of the variable from the industry-specific means. There are 396 distinct four-digit SIC industry dummies. The reported R^2 includes the explanatory power of the industry dummies. The R^2 is 0.288 if we exclude the explanatory power of the industry dummies. Column II—between industry estimates. The coefficients are estimated based on differences between industry-specific means. Column III—within firm estimates. The coefficients are estimated based on variation of the variable from the firm-specific means. There are 9742 distinct firms. The reported R^2 includes the explanatory power of the firm dummies. The R^2 is 0.266 if we do not include the explanatory power of the firm dummies. Column IV—between firm estimates. The coefficients are estimated based on differences between firm-

Table 5: Determinants of market leverage: panel data estimation.

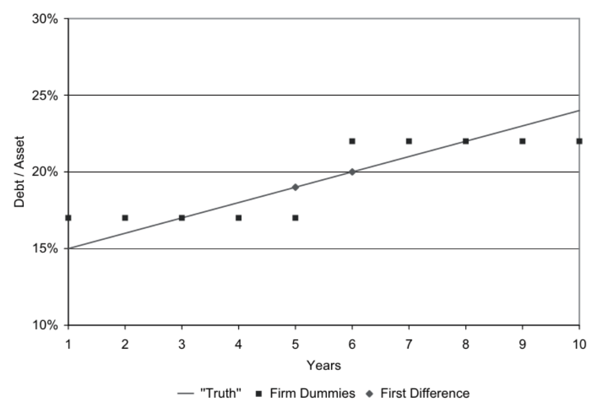

Figure 5
Illustration of panel data estimates
This figure is an illustration of the relative magnitudes of the within and the first-difference estimates of the rating coefficient in a panel data set. In this illustration, the firm's desired but unobserved leverage rises 1% per year over the sample period (the straight line). In the sixth year of the sample, the firm obtains a bond rating and maintains it for the rest of the sample period. The within estimate (like column IV of Table 4) is the difference between the average debt ratio in years when the firm had a debt rating (years 6–10) and years in which it did not (years 1–5). These averages are reported as squares and the difference in the averages is 5% (22.5 – 17.5). The difference coefficient is the difference between the debt

Figure 5: Illustration of panel data estimates.

6.6 Table 6 and Table 7: Interest coverage and first-stage determinants of access

Read: Table 6 uses log interest coverage as the dependent variable and finds rated firms have much lower coverage, consistent with higher debt burden. Table 7 estimates probit first stages for debt ratings. S&P 500 inclusion, NYSE listing, industry rating prevalence, and the minimum-bond-issue-size proxy predict access.

Interpretation: Table 6 shows the leverage result is not just a debt-to-assets artifact. Table 7 shows the instruments are relevant and economically meaningful predictors of rated debt market access.

Economic message: Firms with public debt market access carry more debt service burden, while access itself is predictable from visibility, scale, and industry rating environments.

Table 6
Determinants of interest coverage: firm characteristics

	I	II	III	IV	V
Firm has a debt rating (1 = yes)	−0.650* (0.015)	−0.560* (0.016)	−0.646* (0.015)	−0.586* (0.016)	−0.658* (0.016)
Ln(market assets)	0.105* (0.003)	0.019* (0.004)	0.137* (0.004)	0.144* (0.004)	0.147* (0.004)
Ln(1 + firm age)	0.127* (0.005)	0.111* (0.006)	0.148* (0.005)	0.145* (0.005)	0.150* (0.006)
Profits/sales	5.639* (0.040)	4.406* (0.045)	5.554* (0.041)	5.591* (0.041)	6.145* (0.042)
Tangible assets	−1.345* (0.025)	−0.914* (0.026)	−1.316* (0.025)	−1.261* (0.025)	−1.399* (0.026)
Market-to-book (assets)	0.150* (0.005)	0.225* (0.005)	0.090* (0.005)	0.086* (0.005)	0.069* (0.006)
R&D/sales	−0.810* (0.084)	−0.169** (0.089)	−1.308* (0.089)	−1.307* (0.089)	−1.677* (0.093)
Advertising/sales	−0.768* (0.176)	−0.555* (0.187)	−0.783* (0.178)	−0.729* (0.178)	−0.941* (0.187)
Marginal tax rate		5.011* (0.061)			
Stock return previous year			0.105* (0.009)	0.110* (0.009)	0.137* (0.010)
σ (asset return)			0.800* (0.030)	0.755* (0.030)	0.702* (0.031)
% of debt due in $\leq$ 1 year				0.018 (0.021)	
% of debt due in >5 years				−0.323* (0.019)	
Number of observations	60,701	47,063	57,127	57,127	57,127
Censored observations (%)	17.4	15.9	17.0	17.0	15.1
Pseudo-R^2	0.181	0.214	0.187	0.189	0.195

The dependent variable is the natural log of 1 plus the interest coverage ratio. Interest coverage is operating earnings before depreciation divided by interest expense. The dependent variable is re-coded to zero for observations with negative earnings; the model is then estimated as a tobit with a lower limit of zero (which corresponds to interest coverage of zero), except in column V. In column V, we used a lower limit of –0.69 which corresponds to interest coverage of –0.5 [–0.69 = Ln(1 – 0.5)]. The percent of observations that are censored also are reported in the table. White heteroscedastic consistent errors, corrected for correlation across observations of a given firm, are reported in parentheses [Rogers (1993) and White (1980)]. All models also include year dummy variables and a dummy variable for the regulated utility industry (4900–4939). The sample is based on firms from Compustat that report sales and assets above $1 million between 1986 and 2000 and only includes firm-years with debt. * indicates statistical

Table 6: Determinants of interest coverage.

Table 7
Determinants of bond market access (first stage of instrumental variable regression)

	I	II	III	IV	V	VI
Firm is in the S&P 500		0.550*	0.555*	0.551*	0.562*	0.599*
		(0.081)	(0.081)	(0.081)	(0.081)	(0.080)
Firm trades on the NYSE		0.134*	0.137*	0.135*	0.139*	0.121*
		(0.044)	(0.044)	(0.044)	(0.044)	(0.043)
Log[1 + Pr(rating)] (% of other firms in industry)			0.300***		0.308**	0.324**
			(0.156)		(0.155)	(0.155)
Log[1 + Pr(rating)] (% of other assets in industry)				0.128		
				(0.099)		
Firm is young (age $\leq$ 3)					−0.076	−0.071
					(0.048)	(.048)
Firm is small (18.3% MV asset < Leh min)						−0.425*
						(0.049)
Ln(market assets)	0.547*	0.490*	0.484*	0.488*	0.485*	0.405*
	(0.018)	(0.019)	(0.019)	(0.019)	(0.019)	(0.022)
Ln(1 + firm age)	0.132*	0.075*	0.076*	0.075*	0.051***	0.056**
	(0.017)	(0.018)	(0.018)	(0.018)	(0.027)	(0.027)
Profits/sales	−0.240*	−0.242*	−0.220**	−0.241*	−0.224**	−0.220**
	(0.090)	(0.088)	(0.088)	(0.088)	(0.088)	(0.089)
Tangible assets	0.168**	0.165**	0.127	0.153***	0.124	0.120
	(0.084)	(0.083)	(0.084)	(0.084)	(0.084)	(0.084)
Market-to-book (assets)	−0.153*	−0.158*	−0.155*	−0.157*	−0.155*	−0.161*
	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)
Advertising/sales	0.781**	0.619	0.634***	0.595	0.643***	0.635
	(0.379)	(0.383)	(0.380)	(0.385)	(0.381)	(0.387)
σ (asset return)	−1.730*	−1.787*	−1.743*	−1.781*	−1.747*	−1.751*
	(0.126)	(0.128)	(0.129)	(0.128)	(0.130)	(0.131)
Number of observations	59,562	59,562	59,562	59,558	59,562	59,562
Pseudo-R^2	0.459	0.466	0.466	0.466	0.466	0.471
F-statistic ($\beta_{\text{instruments}} = 0$)		54.3*	59.0*	57.6*	61.5*	125.8*

The table contains estimates from a probit model where the dependent variable is whether the firm has a bond rating (access to the public debt markets). Positive coefficients imply that increases in the variable are associated with a higher probability of a bond rating. White heteroscedastic consistent errors, corrected for correlation across observations of a given firm, are reported in parentheses [Rogers

Table 7: Determinants of bond market access, first stage.

6.7 Table 8: IV determinants of market leverage

Read: Table 8 reports second-stage IV estimates. Across columns, the coefficient on having a debt rating remains around 5.7–6.5 percentage points and statistically significant.

Interpretation: IV lowers the estimated coefficient relative to the OLS baseline but leaves it economically large. This supports the view that bond-market access increases leverage, although it relies on instrument validity.

Economic message: The IV evidence is the paper's strongest attempt to separate supply-side access from unobserved demand for debt. Even predicted access raises leverage materially.

Table 8
Determinants of market leverage (second stage of instrumental variable regression)

	I	II	III	IV	V	VI	VII
Firm has a debt rating	0.078*	0.059*	0.061*	0.060*	0.063*	0.057*	0.065*
(1 = yes)	(0.004)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.010)
Ln(market assets)	−0.025*	−0.023*	−0.023*	−0.023*	−0.023*	−0.023*	−0.020*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Ln(1 + firm age)	−0.016*	−0.016*	−0.016*	−0.016*	−0.016*	−0.015*	−0.016*
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Profits/sales	−0.073*	−0.075*	−0.074*	−0.075*	−0.074*	−0.075*	−0.080*
	(.006)	(.006)	(.006)	(.006)	(.006)	(.006)	(.006)
Tangible assets	0.129*	0.130*	0.130*	0.130*	0.130*	0.130*	0.142*
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Market-to-book (assets)	−0.020*	−0.020*	−0.020*	−0.020*	−0.020*	−0.020*	−0.016*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
R&D/sales	−0.079*	−0.081*	−0.081*	−0.081*	−0.081*	−0.082*	−0.089*
	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)
Advertising/sales	−0.036**	−0.035	−0.035	−0.035	−0.035	−0.035	−0.058*
	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.019)
Stock return	−0.006*	−0.007*	−0.007*	−0.007*	−0.007*	−0.007*	−0.005*
previous year	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
σ (asset return)	−0.334*	−0.334*	−0.334*	−0.334*	−0.334*	−0.334*	−0.322*
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)
Number of observations	59,562	59,562	59,562	59,558	59,562	59,562	66,537
R^2	0.368	0.367	0.367	0.367	0.367	0.367	0.373
Estimation method	OLS	IV	IV	IV	IV	IV	IV

The table contains second stage, instrumental variable estimates, except for column I. In column I, the OLS estimates from Table 4, column I are reproduced. The instruments used in each column (II–VI) are the ones used in the same column of Table 7 (II–VI). In column VII, we use the coefficients from column VI of Table 7 to predict the probability of having a rating for all firms, not just those with positive debt. We then include the firms with zero debt in the second-stage IV estimation. The instruments are: (i) whether the firm is in the S&P 500 (0 or 1); (ii) whether the firm's equity trades on the NYSE (0 or 1); (iii) log of 1 plus the percentage of firms in the same three-digit SIC industry that have a bond rating; (iv) log of 1 plus the percentage of firms in the same three-digit SIC industry that have a bond rating weighted by the market value (MV) of assets; (v) whether the firm's age is three years or less (0 or 1); and (vi) whether

Table 8: Determinants of market leverage, second stage of instrumental-variable regression.

7 Interpretive synthesis

- **Central mechanism.** Firms without access to arm's-length public debt markets rely on private lenders, which may monitor more intensively, charge higher prices, or ration debt quantities.
- **Why rating matters.** A rating is a market-access state variable: it signals that a firm can borrow from public markets and likely issue larger, longer-maturity debt.
- **Demand versus supply.** Rated firms are larger and safer, but those characteristics do not explain the entire leverage gap.
- **Credit constraint interpretation.** The paper is strongest on debt constraints. It is more cautious about full capital constraints because unrated firms may substitute equity for debt.
- **Why this changed the literature.** The paper made debt market access a standard empirical control in corporate finance, especially in studies of financial constraints and capital structure.

8 Short summary

- The paper asks whether firms' access to public debt markets affects leverage.
- Bond/commercial-paper rating is used as a proxy for access.
- Only about 19% of firms have ratings each year, but rated firms issue most of the debt dollars.
- Rated firms have much higher raw leverage and longer debt maturity.
- Rated firms differ substantially in size, profitability, tangibility, growth opportunities, and volatility.

- After controlling for these characteristics, rating status still raises market leverage by about 7–8 percentage points.
- Firm fixed effects leave a positive access effect of about 5 percentage points.
- Interest coverage tests confirm that rated firms carry more debt burden.
- IV estimates using access predictors still find a positive leverage effect of about 5.7–6.3 percentage points.
- The conclusion is that debt supply and public market access matter for capital structure.

9 Conclusion

9.1 Core conclusion

Faulkender and Petersen show that capital structure cannot be understood solely through firms' demand for debt. Firms with access to public debt markets have substantially higher leverage, even after controlling for standard determinants of desired debt and instrumenting for rating status.

9.2 Economic intuition

Information asymmetry and monitoring costs segment debt markets. Public debt markets provide cheaper or more abundant capital for firms that can access them. Firms without access may want more debt but cannot obtain it at the same price or quantity. Thus, observed leverage is jointly determined by demand for debt and supply of debt.

9.3 Incremental contribution in one sentence

The paper's distinctive contribution is to introduce bond-market access as a measurable supply-side determinant of leverage and to show that it has first-order explanatory power in capital structure regressions.

9.4 Final takeaway

The empirical capital structure equation should include capital access. A firm's observed leverage is not just what it wants; it is also what the debt market allows.